

Curriculum Vitae – Romain Pinquie

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I am an Associate Professor in Computer-Mediated Collaborative Engineering Systems Design at the School of Industrial Engineering of Grenoble Institute of Technology and a research fellow at the G-SCOP UMR CNRS Laboratory of Design, Optimisation and Production. I conduct research at the crossroads of systems engineering and human-computer interaction. As a contribution to theory, I try to understand how engineers interact with different types of digital design information, especially for architecting engineered systems. As a contribution to practice, I invent collaborative, highly interactive and visually rich 3D systems modelling user interfaces by parsimoniously mixing three interaction paradigms: Computer-as-tool (HCI) to extend engineers' capabilities, Computer-as-partner (AI) to delegate tasks to artificial agents, and Computer-as-medium (CSCW) to communicate with teammates.

Keywords: *Digital Engineering; Systems Architecting; Model-Based Systems Architecting; Human-Computer Interaction; Human-Systems Integration; Human-Centered Computing; Engineering Systems Design; Computer-Mediated Collaboration; New Product Development; Computer Supported Cooperative Work; Systems Engineering; Model-Based Systems Engineering; Human-Centered Design; 3D User Interface; Virtual Reality; Machine Learning; Natural Language Processing; Text Mining; Conversational AI; Knowledge Engineering.*

EDUCATION

2016 – Arts & Métiers ParisTech, LSIS UMR CNRS 7296, France – Ph.D in Product Design
2012 – Cranfield University, School of Engineering, UK - M.Sc.in Computational Techniques in Engineering
2012 – ESTIA Institute of Technology, France – 2012, M.Eng in Computer-Aided Engineering
2009 – Preparatory classes for the national entrance examination to French engineering schools
2006 – Toulouse 3 University, France – B.Sc. in Mechanical Engineering for Aerospace Technology

POSITIONS HELD

2018 (Fall) - present – Associate professor, Grenoble Institute of Technology, France
2018 (Winter) - 2018 (Fall) – Assistant professor, Arts & Métiers ParisTech – LSIS UMR CNRS, France
2016 - 2017 (Fall) – Post-doctoral research fellow, Airbus Helicopters, France
2013 (Fall) - 2016 (Fall) – Doctoral research fellow, Arts & Métiers ParisTech – LSIS UMR CNRS, France
2013 (Winter) - 2013 (Fall) – Research engineer, Supméca – LISMMMA, France

RESEARCH

STUDENTS

Ph.D. students (245 % of supervision of defended PhD theses without considering resignations):

- **Ghislain Mugisha (2024-2027)**
 - **Supervisors: Pinquie (INPG) 50 % - Loup-Escande (UPJV) 50 %**

- Thesis: Understanding the influence of visual metaphors on cognitive and transdisciplinary collaborative conceptual co-design activities
- Doctoral school: : I-MEP², Université Grenoble Alpes
- Funding: PEPR eSEMBLE
- **Sergio Camilo Medina Galvis** (2024-2027)
 - **Supervisors: Pinquié (INPG) 70 %** - Noël (INPG) 30 %
 - Thesis: Exploring the potential of the Industrial Metaverse in Systems Architecting: hope beyond the hype?
 - Doctoral school: : I-MEP², Université Grenoble Alpes
 - Funding: ANR LabCom MIMESIS
- **Emilie Perreau** (2023-2026)
 - **Supervisors: Pinquié (INPG) 50 %** - Masclet (INPG) 50 %
 - Thesis: Ergonomics guidelines for specifying, designing, and evaluating interactions between a generative conversational AI and a systems engineer
 - Doctoral school: : I-MEP², Université Grenoble Alpes
 - Funding: French Aeronautics and Space Research Foundation
- **Valentin Jousseume** (2022-2025)
 - **Supervisors: Pinquié (INPG) 30 %** - Segonds (ENSAM) 70 %
 - Thesis: Digital adoption of engineering software: application to product lifecycle management
 - Doctoral school: SMI 442, Arts et Métiers ParisTech
 - Funding: French National Research Agency CIFRE project – partnership with KNOWMORE
- **Chao Zhang** (2021-2024)
 - **Supervisors: Pinquié (INPG) 30 %** - Paulette (ENSAM) 30% - Pernot (ENSAM) 40%
 - Thesis: Deep learning for computer-aided reverse engineering of 3D parametric CAD models from 2D technical drawings
 - Doctoral school: SMI 442, Arts et Métiers ParisTech
 - Funding: China Scholarship Council
- **Insaf Nahri** (2021-2024)
 - **Supervisors: Pinquié (INPG) 30 %** - Véron (ENSAM) 70 %
 - Thesis: Artificial intelligence for extracting textual requirements from unstructured building specification documents and integration of formal requirements in a Building Information Modelling digital mock-up
 - Doctoral school: SMI 442, Arts et Métiers ParisTech
 - Funding: French National Research Agency CIFRE project – partnership with CSTB
- **Abdelhadi Lammini** (2021-2024) → *Supervision was discontinued midway through the third year due to concerns about the candidate's non-compliance with scientific integrity and ethical standards. (moved to Antibes and started a startup with INRIA Studio without informing us).*
 - **Supervisors: Pinquié (INPG) 40 %** - Foucault (INPG) 40% - Noël (INPG) 20%
 - Thesis: Mixed Reality for maintaining the geometrical coherence of a Digital Twin
 - Doctoral school : I-MEP², Université Grenoble Alpes
 - Funding: French National Research Agency project – ANR PRC COHERENCE 4D.
- **Haobo Wang** (2021-2024)
 - **Supervisors: Pinquié (INPG) 75 %** - Noël (INPG) 25 %
 - Thesis: Exploring the potential of virtual reality for model-based systems architecting
 - Doctoral school : I-MEP², Université Grenoble Alpes
 - Funding: French National Research Agency project – ANR JCJC ArchiTOOL
- **Victor Romero** (2019-2022)

- Supervisors: **Pinquié (INPG) 50 %** - Noël (INPG) 50 %
- Thesis: An immersive multi-view virtual environment for model-based design reviews
- Doctoral school: I-MEP², Grenoble Alpes University
- Funding: Scholarship from the I-MEP² doctoral school of Grenoble Alpes University
- Award: Best PhD Thesis 2022-2023 Association Française d'Ingénierie Système
- **Armand Huet (2018-2022)**
 - Supervisors: **Pinquié (INPG) 30 %** - Véron (ENSAM) 30 % - Segonds (ENSAM) 40 %
 - Thesis: A context-aware cognitive design assistant
 - Doctoral school: SMI 442, Arts et Métiers ParisTech
 - Funding: Industrial contract, Capgemini
- **Simon Debord (2017-2018 *resignation*)**
 - Supervisors: **Pinquié (INPG) 30 %** - Véron (ENSAM) 30 % - Segonds (ENSAM) 40 %
 - Thesis: Research on a design rules framework
 - Doctoral school: Sciences des Métiers de l'Ingénieur 442, Arts et Métiers ParisTech.
 - Funding: Industrial contract, Capgemini

Post-Doctoral research fellows:

- **Haobo Wang (2024-2027)**
 - Supervisors: **Pinquié (INPG) 100 %**
 - Research topic: A virtual world for the human-centric co-design of a complex system architecture and its industrial system.
 - Funding: ANR LabCom MIMESIS
- **Yaroslav Menshenin (2022-2025)**
 - Supervisors: **Pinquié (INPG) 100 %**
 - Research topic: Model-Based Systems Engineering for the development of new drug-device combination products.
 - Funding: Industrial contract, Becton Dickinson
- **Amer Ezoji (2019-2022)**
 - Supervisors: **Pinquié (INPG) 50 %** - Boujut (INPG) 50 %
 - Research topic: Knowledge capture and reuse for the design of Open Source Hardware products and related services
 - Funding: H2020 Open!Next

M.Sc. Research students:

- **Manon Magendie (2023)**
 - Supervisors : **Pinquié (INPG) 50 %** - Segonds (ENSAM) 50 %
 - Thesis title: AI-based text and free-hand sketch analysis for assessing the additive manufacturability of product concepts during ideation workshops
- **Arthur Glotin (2022)**
 - Supervisors : **Pinquié (INPG) 50 %** - Segonds (ENSAM) 50 %
 - Thesis title: Free-hand sketch analysis for assessing the additive manufacturability of product concepts during ideation workshops
- **Oscar Fossey (2021)**
 - Supervisors: **Pinquié (INPG) 50 %** - Segonds (ENSAM) 50 %
 - Thesis title: Natural language processing for assessing the additive manufacturability of product concepts during ideation workshops
- **Hoa Tran Thi Than (2021)**
 - Supervisors : **Pinquié (INPG) 50%** - Foucault (INPG) 50%

- Thesis title: 3D geometric modelling in virtual reality
- **Victor Romero (2019):**
 - Supervisors: **Pinquié (INPG) 50%** - Noël (INPG) 50%
 - Thesis title: A user-centric computer-aided verification process in a virtuality-reality continuum

PROJECTS

- **Engineering Digital Twins (2026-2029)**
 - **Description:** The [Engineering Digital Twins \(EDT\) research program](#) is a French nationwide initiative, funded by the France 2030 strategy and operated by INRIA, dedicated to advancing the scientific and technical foundations of digital twin technology. The project aims to move beyond ad-hoc solutions by developing standardised, reliable, and interoperable frameworks that bridge the gap between physical entities and their digital counterparts. Organised into five key focus areas – covering hybrid modelling, modular architectures, lifecycle methodologies, real-time synchronisation, and immersive human-digital interaction through AR/VR – the program leverages over 40 diverse use cases and collaborations across major French research institutions to foster a sovereign and innovative digital twin ecosystem.
 - **Scientific coordinators:** Pascale Vicat-Blanc (ENS Lyon) and Benoit Combemale (INRIA).
 - **Funding:** 3-year PhD grant by France 2030.
 - **Role:** PhD co-supervisor with Philippe Palanque (IRIT)
- **BPI Technologies innovantes des univers virtuels immersifs “GAIA XR” (2026-2029)**
 - **Description:** The GAIA XR project aims to develop a collaborative platform for synchronising immersive environments, facilitating the seamless exchange of diverse engineering data throughout a product's lifecycle. By overcoming key technological barriers, the initiative integrates automated 3D CAD analysis, comprehensive MBSE data interoperability, and conflict-free multi-client synchronisation. Furthermore, it pioneers advanced collaborative physics and immersive data visualisation, ensuring that engineering teams can effectively co-design and interact in real-time, whether they are utilising VR headsets or desktop computers. Grenoble INP will focus on integrating diverse MBSE data with digital mock-ups. Furthermore, we will design innovative 3D user interfaces and multimodal metaphors to facilitate MBSE data exploration and collaboration across a variety of hardware in asymmetric collaboration setups.
 - **Scientific coordinator:** Hugo Falgarone, CEO of Skyreal.
 - **Funding:** 166 475 € by BPI France.
 - **Role:** Post-doctoral research fellow supervisor
- **ANR PEPR eSEMBLE (2024-2027)**
 - **Description:** The purpose of the PEPR eSEMBLE (Future of Digital Collaboration) is to fundamentally redefine digital tools for collaboration. To achieve this goal, we must invent mixed (i.e. physical and digital) collaboration spaces that do not simply replicate the physical world in virtual environments, enabling co-located and/or geographically distributed teams to work together smoothly and efficiently. As part of the PEPR eSEMBLE, I obtained a PhD scholarship to understand the influence of visual metaphors on cognitive and transdisciplinary collaborative conceptual co-design activities. The aim of the thesis is threefold. First, the fundamental objective is understanding how interactive visual representations influence the cognitive and collaborative activities of stakeholders involved in co-design activities. Second, a methodological aim is to propose a protocol for evaluating visual metaphors and metrics for measuring cognitive and collaborative activities in a transdisciplinary computer-mediated co-design activity. Third, the final goal is to develop ergonomic recommendations that provide concrete, actionable guidance for designing and evaluating virtual metaphors and ease the adoption of interactive visualisation in collaborative conceptual design phases.

- **Scientific coordinator:** Gilles Bailly (CNRS), Michel Beaudouin-Lafon (Université Paris-Saclay), Laurence Nigay (UGA), Nicolas Roussel (INRIA)
- **Funding:** 3 years PhD scholarship + 10 000 € by the French National Research Agency (ANR).
- **Role:** Ph.D co-supervisor with Émilie Loup-Escande (Université Picardie Jules-Verne, Centre de Recherche en Psychologie : Cognition, Psychisme et Organisation)

▪ **ANR LabCom Joint Research Lab MIMESIS et Institut Carnot LSI (2024-2029)**

- **Description:** The iMmersive huMan-systEms InteractionS lab (MIMESIS'Lab) is a joint research laboratory between the G-SCOP UMR CNRS Laboratory of Design, Optimisation and Production at the Grenoble Institute of Technology (Grenoble INP-UGA) and the software editor SKYREAL. Our research tries to understand how humans represent and interact with different types of design information, and to develop natural and visually rich 3D human-computer interactions to advance human-centred systems engineering. Short-term objectives include the development of new multi-modal collaborative modelling interfaces to explore and manage digital threads and to co-design architectures of engineered systems.
- **Scientific coordinator:** Romain Pinquié, Grenoble Institute of Technology (Grenoble INP).
- **Funding:**
 - 363 000 € by the French National Research Agency (ANR).
 - 56 000 € by Institut Carnot LSI in 2024
 - 71 000 € by Institut Carnot LSI in 2025
- **Role:** Project scientific coordinator and Ph.D co-supervisor

▪ **AI-Driven Engineering Assistance System (AIDEAS) (2023-2026)**

- **Description:** The objectives of the AIDEAS joint research project between the technological research institute Saint-Exupéry and Grenoble Institute of Technology are twofold. First, it aims to tune Large Language Models to tailor generative conversational AI to systems engineering activities. Second, the objective is to define methodological ergonomics guidelines for specifying, designing, implementing and evaluating interactions between a systems engineer and a conversational generative AI.
- **Coordinator:** Romain Pinquié, Grenoble Institute of Technology (Grenoble INP).
- **Funding:** 546 749 € by the French Aeronautics and Space Research Foundation.
- **Role:** Project scientific coordinator and Ph.D co-supervisor

▪ **Industrial contract Becton Dickinson (2022-2025)**

- **Description:** This action research project will integrate various methodological engineering design elements into a model-based systems engineering framework (process, method, modelling method, modelling language, modelling software) to support the Advanced Drug Delivery Systems division of Becton Dickinson Medical Pharmaceutical Systems in the development of future new MedTech drug-device combination products.
- **Scientific coordinator:** Romain Pinquié, Grenoble Institute of Technology (Grenoble INP).
- **Funding:** 285 000€ by Becton Dickinson – Medical Pharmaceutical Systems.
- **Role:** Project scientific coordinator

▪ **ANR EQUIPEX+ Continuum (2021-2031)**

- **Description:** The CONTINUUM research project will create a collaborative research infrastructure of 30 platforms located throughout France, to advance interdisciplinary research based on interactions between computer science and the human and social sciences. Thanks to CONTINUUM, 37 research teams will develop cutting-edge research programs focusing on visualization, immersion, interaction

and collaboration, as well as on human perception, cognition and behaviour in virtual/augmented reality. ([Website](#))

- **Scientific coordinator:** Michel Beaudouin-Lafon, French National Research Council (CNRS).
- **Funding:** 13.6M€ by Investments of the Future national program.
- **Role:** Member of the scientific committee and of the technology transfer committee

▪ **ANR PRC 2020 “Coherence4D” ANR-20-CE10-0002 (2021-2024)**

- **Description:** The COHERENCE 4D research project will develop a new paradigm of modelling, visualisation, interaction, and coherence maintenance of the digital twin interfaced with the physical twin. Thus, the maintenance of coherence will be done in four dimensions to take into account the spatio-temporal character of the evolutions: the digital twin modelled in three dimensions (3D) adapts to the temporal evolutions (4th dimension) of the physical twin. ([Website](#))
- **Scientific coordinator:** Jean-Philippe Pernot, Arts et Métiers ParisTech.
- **Funding:** 623 188€ by the French National Research Agency (ANR).
- **Role:**
 - Scientific coordinator of the work package 5 “integration and validation”
 - Ph.D co-supervisor

▪ **ANR JCJC 2020 “ArchiTool” ANR-20-CE10-0010 (2021-2024)**

- **Description:** The ArchiTOOL research project aims at inventing, prototyping, and evaluating an immersive and intelligent virtual environment for collaboratively designing the architecture of engineered systems. Instead of using domain-specific engineering software, the immersive and interactive environment will provide the architect with the modelling capabilities required to define the various views (operational, specification, functional, behavioural, structural, logic, safety, etc.) of a system architecture in a single virtual space before exporting each viewpoint in a standardised format that will enable domain-experts to continue with a detailed design. Moreover, the immersive environment will include a cognitive agent to support the system architect with intelligent capabilities: model verification, context-aware recommendation of rules, identification and automation of modelling routines. ([Website](#))
- **Scientific coordinator:** Romain Pingué, Grenoble Institute of Technology (Grenoble INP).
- **Funding:** 172 502€ by the French National Research Agency (ANR).
- **Role:** Project scientific coordinator and Ph.D. co-supervisor

▪ **Horizon 2020 “Open!Next” (2019-2022)**

- **Description:** The OPEN!NEXT innovation project aims to enable SMEs to engage in company-community collaboration for the co-development and market exploitation of open source hardware (OSH) products and related services. OPEN!NEXT builds upon the vision that SMEs are the best placed to release the formidable potential of OSH in terms of product innovation and business incubation. The project aims to establish a company-community ecosystem, facilitate the engagement of SMEs in efficient collaboration with OSH communities and open up new avenues for value creation. ([Website](#))
- **Scientific coordinator:** Robert Mies, Technische Universität Berlin.
- **Funding:** 5,8M€ by the European Union’s Horizon 2020 research and innovation programme.
- **Role:** Member of the steering committee and postdoctoral research fellow co-supervisor

PUBLICATIONS

Theses

[1] R. Piquié. (2016) **A requirement mining framework**. Ph.D. Thesis. Arts & Métiers ParisTech. LSIS UMR CNRS 7296 Laboratory. Aix-en-Provence, France.

[2] R. Piquié. (2012) **Prognostics and health management design technology: research on the application of functional modelling to structural integrity problems**. M.Sc. Thesis. Cranfield University. School of Engineering. Applied Mathematics & Computing Group. Cranfield, United Kingdom.

Refereed Journal Articles (16 publications)

--- 2026 ---

[16] H. Wang, **R. Piquié**, F. Noel (2026). **Comparing diagrams with an immersive virtual world for visual conceptual modelling in systems engineering: an exploratory mixed-methods study**. International Journal of Human-Computer Interaction. (SJIR 2024=1.383 Q1, H-Index=110, IF=4.9)
DOI: [... \(in press\)](#)

[15] C. Zhang, A. **R. Piquié**, Polette, M. Iida, H. De Charnace, J-P. Pernot (2026). **3D CAD Models Reconstruction from 2D Drawings: A Survey and Future Challenges**. Computer-Aided Design and Applications (CADA), Vol. 23, No. 6, pp. 646-678, 2026.
DOI: [10.14733/cadaps.2026.646-678](#)

[14] H. Wang, **R. Piquié**, F. Noel (2026). **An immersive virtual world for co-architecting engineered systems**. Virtual Reality. (SJIR 2024=1.198 Q1, H-Index=73, IF=5.0)
DOI: [10.1007/s10055-025-01302-1](#)

[13] A. Lammini, F. Noël, G. Foucault, **R. Piquié** (2026) **An augmented reality framework for real-time positioning and orientation synchronization of CAD models with physical environments: application to digital twin coherence**. Int J Adv Manuf Technol. (SJIR 2024=0.706 Q1, H-Index=175, IF=3.1)
DOI: [10.1007/s00170-026-17766-7](#)

[12] V. Jousseume, **R. Piquié**, F. Frayssé, E. Esquieu, F. Segonds (2026). **A Comparison of the Effects of a Digital Adoption Platform on Product Lifecycle Management Software**. Journal of Internet Services and Information Security (JISIS), 16(1), 160-180. (SJIR 2024=0.395 Q2, H-Index=17)

--- 2025 ---

[11] C. Zhang, A. Polette, **R. Piquié**, M. Iida, H. De Charnace, J-P. Pernot (2025) **Reinforcement Learning-Based Parametric CAD Models Reconstruction From 2D Orthographic Drawings**. In *Computer Aided Design*, Vol. 188, 2025. (SJIR 2024=0.766 Q1, H-Index=133, IF=3.1)
DOI: [10.1016/j.cad.2025.103925](#)

[10] C. Zhang, A. Polette, **R. Piquié**, G. Carasi, H. De Charnace, J-P. Pernot (2025) **eCAD-Net: editable parametric CAD models reconstruction from dumb B-Rep models using deep neural networks**. In *Computer-Aided Design*, Vol. 118, 2025. (SJIR 2024=0.766 Q1, H-Index=133, IF=3.1)
DOI: [10.1016/j.cad.2024.103806](#)

--- 2023 ---

[9] C. Zhang, **R. Piquié**, A. Polette, G. Carasi, H. De Charnace, J-P. Pernot (2023) **Automatic 3D CAD models reconstruction from 2D orthographic drawings**. In *Computers & Graphics, Special Issue on Shape Modeling International (SMI) 2023*, Vol. 114, pp. 179-189, 2023. (SJIR 2024=0.569 Q2, H-Index=88, IF=2.8)
DOI: [10.1016/j.cag.2023.05.021](#)

--- 2022 ---

[8] **R. Piquié**, O. Fossey, F. Segonds (2022) **DREAM: a design assistant for assessing additive manufacturability**. In *International Journal of Product Lifecycle Management (IJPLM)*, Vol. 14, No. 5, pp. 328-349, 2022. (SJIR 2024=0.219 Q3, H-Index=25)
DOI: [10.14733/cadaps.2022.1184-1190](#)

[7] V. Romero, **R. Piquié**, F. Noël (2022) **A user-centric computer-aided verification process in a virtuality-**

reality continuum. In *Computers in Industry*, Vol. 140, 2022. (SJR 2024=2.209 Q1, H-Index=139, IF=9.1)
DOI: [10.1016/j.compind.2022.103678](https://doi.org/10.1016/j.compind.2022.103678)

[6] T.T.H. Tran, G. Foucault, **R. Pinqu  ** (2022) **Benchmarking of 3D modelling in virtual reality.** In *Computer-Aided Design and Applications (CADA)*, Vol. 19, No. 6, pp. 1184-1190, 2022.
DOI: [10.14733/cadaps.2022.1184-1190](https://doi.org/10.14733/cadaps.2022.1184-1190)

[5] A. Huet, **R. Pinqu  **, P. V  ron, F. Segonds, V. Fau (2022) **Design rules application in manufacturing industries: a state-of-the-art survey and proposal of a context-aware approach.** In *International Journal on Interactive Design and Manufacturing (IJIDeM)*, Vol. 16, pp. 317-322, 2022. (SJR 2024=0.493 Q2, H-Index=44, IF=2.5)
DOI: [10.1007/s12008-021-00821-w](https://doi.org/10.1007/s12008-021-00821-w)

--- 2021 ---

[4] A. Huet, F. Segonds, **R. Pinqu  **, P. V  ron, J. Guegan, A. Mallet (2021) **Context-aware cognitive design assistant: Implementation and study of design rules recommendations.** In *Advanced Engineering Informatics*, Vol. 50, 2021. (SJR 2024=1.993 Q1, H-Index=124, IF=9.9)
DOI: [10.1016/j.compind.2020.103377](https://doi.org/10.1016/j.compind.2020.103377)

--- 2020 ---

[3] A. Huet, **R. Pinqu  **, P. V  ron, A. Mallet, F. Segonds, N. Crou   (2020) **CACDA: A knowledge graph for a context-aware cognitive design assistant.** In *Computers in Industry*, Vol. 125, 2020. (SJR 2024=2.209 Q1, H-Index=139, IF=9.1)
DOI: [10.1016/j.compind.2020.103377](https://doi.org/10.1016/j.compind.2020.103377)

--- 2016 ---

[2] **R. Pinqu  **, P. V  ron, F. Segonds, N. Crou   (2016) **Requirement mining for model-based product design.** In *International Journal of Product Lifecycle Management*, Vol. 9, No. 4, pp. 305-332, 2016. (SJR 2024=0.219 Q3, H-Index=25)
DOI: [10.1504/IJPLM.2016.080983](https://doi.org/10.1504/IJPLM.2016.080983)

--- 2015 ---

[1] **R. Pinqu  **, L. Rivest, F. Segonds, P. V  ron **An illustrated glossary of ambiguous terms used in discrete manufacturing.** In *International Journal of Product Lifecycle Management*, Vol. 8, No. 2, pp. 142-171, 2015. (SJR 2024=0.219 Q3, H-Index=25)
DOI: [10.1504/IJPLM.2015.070580](https://doi.org/10.1504/IJPLM.2015.070580)

Refereed International Conference Papers (28 publications)

--- 2026 ---

[35] N. Neumann, **R. Pinqu  **, A. Geovana Blayer Ribeiro (2026) **Mapping design research methods: foundations for a design research quality hub.** 19th International Design Conference, Dubrovnik-Cavtat, Croatia, May 18-21, 2026.
DOI: *in press*

[34] **R. Pinqu  **, S. C. M., Galvis, H. Wang, G. Mugisha, G. (2026) **Talk, Touch, See, Explore: A Multimodal Virtual Environment for Digital Thread Analytics with a Conversational Generative AI Assistant.** In *2026 International Workshop on Systems Engineering Research* (pp. 1-8). IEEE.
DOI: [10.1109/IEEECONF67861.2026.11440866](https://doi.org/10.1109/IEEECONF67861.2026.11440866)

[33] E. Perreau, Y. Menshenin, **R. Pinqu  ** (2026) **Human-systems integration through the lenses of human factors ergonomics and systems engineering: from cooperation to collaboration.** In *2026 International Workshop on Systems Engineering Research* (pp. 1-8). IEEE.
DOI: [10.1109/IEEECONF67861.2026.11440740](https://doi.org/10.1109/IEEECONF67861.2026.11440740)

--- 2025 ---

[32] E. Perreau, **R. Pinqu  **, R. Drouin, J. Huchard, J. D. Piques (2025). **Collective Reflection on Conversational Generative AI for (Model-Based) Systems Engineering: Insights from the INCOSE French Chapter.** In *2025 IEEE International Symposium on Systems Engineering (ISSE)* (pp. 1-8). IEEE.
DOI: [10.1109/ISSE65546.2025.11370089](https://doi.org/10.1109/ISSE65546.2025.11370089)

- [31] Y. Menshenin, **R. Pinquié** (2025). **How Does Modeling Humans as System Elements Influence Architectural Trade-Off Decisions?**. In *2025 IEEE International Symposium on Systems Engineering (ISSE)* (pp. 1-8). IEEE.
DOI: [0.1109/ISSE65546.2025.11370059](https://doi.org/0.1109/ISSE65546.2025.11370059)
- [30] Y. Menshenin, J. Cancellieri, G. Bonnefond, **R. Pinquié**, P. Chévrier (2025). **Holistic Model-Based Perspective on MedTech-Pharma Interfaces for New Combination Product Development**. In *The 27th International DSM Conference (DSM 2025)* (pp. 1-10).
DOI: [10.35199/dsm2025.01](https://doi.org/10.35199/dsm2025.01)
- [29] V. Jousseau, F. Fraysse, E. Esquieu, **R. Pinquié**, F. Segonds (2025). **Methodological Proposal for the Deployment of PLM Software Supported by a Digital Adoption Platform and AI Models**. In *IFIP International Conference on Product Lifecycle Management* (pp. 141-150). Cham: Springer Nature Switzerland.
DOI: *in press*
- [28] S. C. Medina Galvis, **R. Pinquié**, B. Paterné (2025) **Exploring Model-Based Systems Architecting Digital Threads in Virtual Reality**. IFIP 21st International Conference on Product Lifecycle Management (PLM), Bangkok, Thailand, July 7-10, 2024. In: Sureephong, P., Danjou, C., Bouras, A. (eds) Product Lifecycle Management. Integrating Digital Technologies for Sustainability and Innovation. PLM 2024. IFIP Advances in Information and Communication Technology, vol 740. Springer, Cham.
DOI: [10.1007/978-3-031-93319-6_5](https://doi.org/10.1007/978-3-031-93319-6_5)
- [27] I. Nahri, **R. Pinquié**, P. Véron, N. Bus, M. Thorel (2025) **Named Entity Recognition for Building Information Model Verification using Large Language Models**. IFIP 21th International Conference on Product Lifecycle Management (PLM), Bangkok, Thailand, July 7-10, 2024. In: Sureephong, P., Danjou, C., Bouras, A. (eds) Product Lifecycle Management. Integrating Digital Technologies for Sustainability and Innovation. PLM 2024. IFIP Advances in Information and Communication Technology, vol 740. Springer, Cham.
DOI: [10.1007/978-3-031-93319-6_33](https://doi.org/10.1007/978-3-031-93319-6_33)
- 2024 ---
- [26] E. Perreau, **R. Pinquié**, C. Masclet (2024) **A Pilot Study for Analyzing Systems Engineer-Conversational GenAI Interaction: A Case Study for Requirements Development & Validation**. 3rd International Conference on Human Systems Integration, Jeju, Korea, August 27-29, 2024.
- [25] Y. Menshenin, **R. Pinquié** (2024) **The Influence of Human-Systems Integration on the “Fuzzy Front End” of Innovation: A MedTech Case Study**. 3rd International Conference on Human Systems Integration, Jeju, Korea, August 27-29, 2024.
- [24] T. Thanh Hoa Tran, G. Foucault, **R. Pinquié** (2024) **Direct Solid Modelling in Virtual Reality for Systems Architecting**. 34th CIRP Design Conference, Cranfield, United Kingdom, June 3-5, 2024. In *Procedia CIRP*. Vol. 128, pp. 656-661, 2024.
DOI: [10.1016/j.procir.2024.08.230](https://doi.org/10.1016/j.procir.2024.08.230)
- [23] **R. Pinquié**, L. Roucoules, P-A. Yvars, R. Chenouard (2024) **Operationalizing community-based open scientific design research benchmarks: application to model-based architecture design synthesis**. 18th International Design Conference, Dubrovnik-Cavtat, Croatia, May 20-24, 2024.
DOI: [10.1017/pds.2024.12](https://doi.org/10.1017/pds.2024.12)
- 2023 ---
- [22] Y. Menshenin, **R. Pinquié**, C. Chevrier (2023) **Defining a new role of system architect at the product strategy stage of a MedTech product development through the DSM-based method**. 25th International DSM Conference (DSM 2023), Gothenburg, Sweden, October 03-05, 2023. In *Proceedings of the Design Society*, pp. 116-122, 2023.
DOI: [10.35199/dsm2023.13](https://doi.org/10.35199/dsm2023.13)
- [21] Y. Menshenin, **R. Pinquié**, P. Chevrier (2023) **Holistic Perspective to the Drug-Device Combination Product Development Challenges**. IFIP 20th International Conference on Product Lifecycle Management (PLM), Montreal,

QC, Canada, July 9-12, 2023. In: Danjou, C., Harik, R., Nyffenegger, F., Rivest, L., Bouras, A. (eds) Product Lifecycle Management. Leveraging Digital Twins, Circular Economy, and Knowledge Management for Sustainable Innovation. PLM 2023. IFIP Advances in Information and Communication Technology, vol 701. Springer, Cham.
DOI: [10.1007/978-3-031-62578-7](https://doi.org/10.1007/978-3-031-62578-7)

[20] Y. Menshenin, **R. Piquié**, P. Chevrier (2023) **New product development process for MedTech combination products**. 24th International Conference on Engineering Design (ICED), Bordeaux, France, July 24-28, 2023. In *Proceedings of the Design Society*, 3, pp. 2795-2804, 2023.
DOI: [10.1017/pds.2023.280](https://doi.org/10.1017/pds.2023.280)

[19] **R. Piquié**, H. Wang, F. Noël (2023) **Human-centric co-design of model-based system architecture**. 33rd CIRP Design Conference, Sydney, Australia, May 17-19, 2023. In *Procedia CIRP*. Vol. 119, pp. 146-151, 2023.
DOI: [10.1016/j.procir.2023.02.131](https://doi.org/10.1016/j.procir.2023.02.131)

[18] A. Huet, **R. Piquié**, F. Segonds, P. Véron (2023) **A cognitive design assistant for context-aware computer-aided design**. 33rd CIRP Design Conference, Sydney, Australia, May 17-19, 2023. In *Procedia CIRP*. Vol. 119, pp. 1029-1034, 2023.
DOI: [10.1016/j.procir.2023.03.146](https://doi.org/10.1016/j.procir.2023.03.146)

--- 2022 ---

[17] V. Romero, **R. Piquié**, F. Noël. (2022) **An open benchmark exercise for model-based design reviews**. IFIP 19th International Conference on Product Lifecycle Management (PLM), Grenoble, France, July 10-13, 2022. In: Noël, F., Nyffenegger, F., Rivest, L., Bouras, A. (eds) Product Lifecycle Management. PLM in Transition Times: The Place of Humans and Transformative Technologies. PLM 2022. IFIP Advances in Information and Communication Technology, vol 667. Springer, Cham.
DOI: [10.1007/978-3-031-25182-5_18](https://doi.org/10.1007/978-3-031-25182-5_18)

[16] A. Lammini, **R. Piquié**, G. Foucault, F. Noël. (2022) **Geometric coherence of a digital twin: a discussion**. IFIP 19th International Conference on Product Lifecycle Management (PLM), Grenoble, France, July 10-13, 2022. In: Noël, F., Nyffenegger, F., Rivest, L., Bouras, A. (eds) Product Lifecycle Management. PLM in Transition Times: The Place of Humans and Transformative Technologies. PLM 2022. IFIP Advances in Information and Communication Technology, vol 667. Springer, Cham. [10.1007/978-3-031-25182-5_23](https://doi.org/10.1007/978-3-031-25182-5_23)

[15] **R. Piquié**, V. Romero, F. Noël. (2022) **Survey of model-based design review: practices and challenges**. 17th International Design Conference, Dubrovnik-Cavtat, Croatia, May 23-26, 2022.
DOI: [10.1007/978-3-031-25182-5_23](https://doi.org/10.1007/978-3-031-25182-5_23)

[14] **R. Piquié**, J. Le Duigou, L. Grimal, L. Roucoules. (2022) **On open science platform for benchmarking engineering design researches**. 32nd CIRP Design Conference, Gif-sur-Yvette, France, March 28-30, 2022. In *Procedia CIRP*. Vol. 109, pp. 472-477, 2022.
DOI: [10.1016/j.procir.2022.05.280](https://doi.org/10.1016/j.procir.2022.05.280)

--- 2021 ---

[13] R. Antoniou, **R. Piquié**, J-F Boujut, A. Ezoji. (2021) **Identifying the factors affecting the replicability of open-source hardware designs**. 23rd International Conference on Engineering Design (ICED), Gothenburg, Sweden, August 16-20, 2021.
DOI: [10.1017/pds.2021.443](https://doi.org/10.1017/pds.2021.443)

[12] V. Romero, **R. Piquié**, F. Noël. (2021) **An immersive virtual environment for reviewing model-centric designs**. 23rd International Conference on Engineering Design (ICED), Gothenburg, Sweden, August 16-20, 2021.
DOI: [10.1017/pds.2021.45](https://doi.org/10.1017/pds.2021.45)

[11] A. Ezoji, J.-F. Boujut, **R. Piquié**. **Requirements for design reuse in open-source hardware: a state of the art**. 31st CIRP Design Conference, Enschede, Netherlands, May 18-21, 2021. In *Procedia CIRP*. Vol. 100, pp. 792-797, 2021.
DOI: [10.1016/j.procir.2021.05.042](https://doi.org/10.1016/j.procir.2021.05.042)

[10] T.T.H. Tranh, G. Foucault, **R. Piquié**. (2021) **Benchmarking of 3D modelling in virtual reality environments**. 21st annual International CAD Conference, Barcelona, Spain, July 5-7, 2021 (selected for the international journal Computer-Aided Design and Applications).
DOI: [10.14733/cadconfP.2021.324-328](https://doi.org/10.14733/cadconfP.2021.324-328)

[9] V. Romero, **R. Piquié**, F. Noël. (2021) **A computer-aided verification process for engineered systems**. Conférence Internationale Génie Industriel (CIGI) QUALITA, Grenoble, France, May 5-7, 2021 (selected for the international journal Computers in Industry).

--- 2020 ---

[8] A. Huet, **R. Piquié**, F. Segonds, V. Fau. (2020) **Knowledge graph of design rules for a context-aware cognitive design assistant**. In: Nyffenegger F., Ríos J., Rivest L., Bouras A. (eds) Product Lifecycle Management Enabling Smart X. PLM 2020. IFIP Advances in Information and Communication Technology, vol 594. Springer, Cham.
DOI: [10.1007/978-3-030-62807-9_27](https://doi.org/10.1007/978-3-030-62807-9_27)

[7] A. Ezoji, **R. Piquié**, J-F. Boujut. (2020) **Towards a better understanding of open source hardware design reuse in company-community collaboration**. In: Roucoules L., Paredes M., Eynard B., Morer Camo P., Rizzi C. (eds) Advances on Mechanics, Design Engineering and Manufacturing III. JCM 2020. Lecture Notes in Mechanical Engineering. Springer, Cham.
DOI: [10.1007/978-3-030-70566-4_24](https://doi.org/10.1007/978-3-030-70566-4_24)

[6] A. Huet, **R. Piquié**, P. Véron, F. Segonds, V. Fau. (2020) **Design rules application in manufacturing industries: a state-of-the-art survey and proposal of a context-aware approach**. In: Roucoules L., Paredes M., Eynard B., Morer Camo P., Rizzi C. (eds) Advances on Mechanics, Design Engineering and Manufacturing III. JCM 2020. Lecture Notes in Mechanical Engineering. Springer, Cham.
DOI: [10.1007/978-3-030-70566-4_53](https://doi.org/10.1007/978-3-030-70566-4_53)

--- 2019 ---

[5] **R. Piquié**, P. Véron, F. Segonds, T. Zynda. (2019) **A property graph data model for a context-aware design assistant**. In: Fortin C., Rivest L., Bernard A., Bouras A. (eds) Product Lifecycle Management in the Digital Twin Era. PLM 2019. IFIP Advances in Information and Communication Technology, vol 565. Springer, Cham.
DOI: [10.1007/978-3-030-42250-9_17](https://doi.org/10.1007/978-3-030-42250-9_17)

--- 2018 ---

[4] **R. Piquié**, P. Véron, F. Segonds, N. Croué. (2018) **A requirement mining framework to support sub-systems suppliers**. 28th CIRP Design Conference, Nantes, France, May 23-25, 2018. In *Procedia CIRP*. Vol. 70, pp. 410-415, 2018 (Nominated for the best paper award).
DOI: [10.1016/j.procir.2018.03.228](https://doi.org/10.1016/j.procir.2018.03.228)

--- 2016 ---

[3] **R. Piquié**, P. Micouin, P. Véron, F. Segonds. (2016) **Property model methodology: a case study with Modelica**. 11th International Tools and Methods of Competitive Engineering (TMCE) symposium, Aix-en-Provence, France, May 9-13, 2016.

[2] **R. Piquié**, P. Véron, F. Segonds, N. Croué. (2016) **Natural Language Processing of Requirements for Model-Based Product Design with ENOVIA/CATIA V6**. In: Bouras, A., Eynard, B., Foufou, S., Thoben, KD. (eds) Product Lifecycle Management in the Era of Internet of Things. PLM 2015. IFIP Advances in Information and Communication Technology, vol 467. Springer, Cham.
DOI: [10.1007/978-3-319-33111-9_19](https://doi.org/10.1007/978-3-319-33111-9_19)

--- 2015 ---

[1] **R. Piquié**, P. Véron, F. Segonds, N. Croué. (2015) **A collaborative requirement mining framework to support OEMs**. In: Luo, Y. (eds) Cooperative Design, Visualization, and Engineering. CDVE 2015. Lecture Notes in Computer Science(), vol 9320. Springer, Cham.
DOI: [10.1007/978-3-319-24132-6_13](https://doi.org/10.1007/978-3-319-24132-6_13)

Book Chapters (2 publications)

Pinquié, R. (2026). Advancing Human-Computing in Model-Based Systems Architecting. In *Handbook of Sociotechnical Systems* (pp. 218-234). CRC Press.

Boujut, J. F., & Pinquié, R. (2024). Building Great Documentation. *Open-Source Hardware Development. A handbook for collaborative product creation*, 72-95.

Refereed National Conference Papers (3 publications)

[3] E. Perreau, R. Pinquié, C. Masclet. (2024) **Proposition d'une méthodologie de recherche pour l'étude des interactions humains-agent conversationnel textuel en milieu écologique : application pour des tâches d'Ingénierie Système**. In ModACT, Paris, France, May 22-24, 2024. (Best presentation award)

[2] A. Glotin, R. Pinquié, A. Ouradi, C. Ouazine, F. Segonds. (2022) **Dream: design rules extractor for additive manufacturability, processing of natural language and free-hand sketches**. In CONFERE, Bâle, Switzerland, July 7-8, 2022.

[1] S. Debord, F. Segonds, R. Pinquié, P. Véron, N. Croué. (2018) **Proposition of design rules framework**. CONFERE, Budapest, Hungary, July 5-7, 2018.

MEMBERSHIP OF COMMITTEES, COMMISSIONS, BOARDS, OR SIMILAR

- Head of Concurrent & Integrated Design Research Group, G-SCOP UMR 5272 laboratory since 2026
- Scientific committee member of the Multidisciplinary Institute in Artificial Intelligence (MIAI Cluster) since 2025
- Elected board member as VP of Research & Education, INCOSE French Chapter AFIS since 2025
- Scientific committee member of GIS S.smart national special interest group since 2023
- Elected board member of the G-SCOP UMR 5272 laboratory since 2023
- Elected scientific committee member of the [IFIP WG 5.1](#) since 2017
- INCOSE French Chapter, board member of the Rhône-Alpes local committee

SCIENTIFIC DUTIES

Review of papers for journals

- Human Computer Interaction
- Virtual Reality
- Journal of Engineering Design
- Research in Engineering Design
- Design Science
- Computer-Aided Design and Applications
- Computers in Industry
- Concurrent Engineering: Research and Applications

Participating in or organising international conference as part of the Program Committee

2026:

- 17th Int. Conference on Complex Systems Design & Management (CSD&M), November 19-20, 2026, Paris, France.

- 1st IEEE-INCOSE Int. Workshop on Systems Engineering Research, January 14, 2026, Toulouse, France. [General Chair]
- 22nd Int. Conference on Product Lifecycle Management 2025, July 7-9, 2025, Sevilla, Spain. [Digital Twin Contest Chair]

2025:

- 23rd Int. Conference on Research and Education in Mechatronics, December 3-5, 2025, Paris, France. [PC member]
- 21st Int. Conference on Product Lifecycle Management 2025, July 7-9, 2025, Sevilla, Spain. [Digital Twin Contest Chair]

2024:

- 20th Int. Conference on Product Lifecycle Management 2024, July 8-10, 2024, Bangkok, Thailand. [Digital Twin Contest Chair]

2022:

- 19th Int. Conference on Product Lifecycle Management 2022, July 10-13, 2022, Grenoble, France. [Scientific Committee Chair]

Review of submissions for International Conferences as Part of the Scientific Committee

2026:

- 19th Int. Design Conference 2026, May 18-21, 2026, Dubrovnik, Croatia.
- 17th Int. Conference on Complex Systems Design & Management (CSD&M), November 19-20, 2026, Paris, France.
- 13th Int. International Joint Conference on Mechanics, Design Engineering and Advanced Manufacturing (JCM), June 17-19, 2026, Aix-les-Bains, France.

2025:

- 36th Conférence Int. Francophone sur l'Interaction Humain-Machine, November 3-7, 2025, Toulouse, France.
- 25th Int. Conference on Engineering Design, August 11-14, 2025, Dallas, Texas, USA.
- 22nd Int. Conference on Product Lifecycle Management 2022, July 7-9, 2025, Sevilla, Spain.
- 13th Int. Conference on Model-Based Software and Systems Engineering, February 26-28, 2025, Porto, Portugal.
- 10th Int. Conference on Research Into Design, January 8-10, 2025, Hyderabad, India.
- 1st Int. AFIS-IEEE Workshop on Systems Engineering Research, January 14, 2025, Toulouse, France.

2024:

- 15th Int. Complex Systems Design & Management Conference, December 12-13, 2024, Paris, France.
- 3rd Int. INCOSE Human-Systems Integration Conference 2024, August 28-30, 2024, Jeju, Korea.
- 20th Int. Conference on Product Lifecycle Management 2022, July 8-10, 2024, Bangkok, Thailand.
- 18th Int. Design Conference 2022, May 20-23, 2024, Dubrovnik, Croatia.
- 12th Int. Conference on Model-Based Software and Systems Engineering, February 21-23, 2024, Rome, Italy.

2023:

- 20th Int. Conference on Product Lifecycle Management 2023, July 9-12, 2023, Montréal, Canada.
- 24th Int. Conference on Engineering Design (ICED22), August 2023, Bordeaux, France.

2022:

- 19th Int. Conference on Product Lifecycle Management 2022, July 10-13, 2022, Grenoble, France.
- 17th Int. Design Conference 2022, May 23-26, 2022, Dubrovnik, Croatia.

2021:

- 23rd Int. Conference on Engineering Design (ICED21), August 2021, Gothenburg, Sweden.
- 18th Int. Conference on Product Lifecycle Management, July 11-14, 2021, Parana, Brazil.

2020:

- 17th Int. Conference on Product Lifecycle Management, July 5-8 2020, Rapperswil, Switzerland.

2019:

- 22nd Int. Conference on Engineering Design (ICED19), August 5-8, 2019, Delft, The Netherlands.
- 16th Int. Conference on Product Lifecycle Management, July 8-12, 2019, Moscow, Russia.
- 16th Int. CAD Conference, June 24-26, 2019, Singapore.

2018:

- 87th INCOSE Great Lake Regional Conference, October 17-20, 2018, Indianapolis, USA.
- 15th Int. CAD Conference, July 9-11, 2018, Paris, France.
- 28th INCOSE Int. Symposium, July 7-12, 2018, Washington, USA.

Review of submissions to calls for funding

- Call "exploratory research projects" (MIAI Institute) 2026.
- Research call from Haute école spécialisée de Suisse occidentale (HES-SO) 2025.
- Specific call "Recherche-Action SIOMRI" of the French National Research Agency (ANR) 2021.
- Generic call 2022 of the French National Research Agency (ANR) 2021.
- Industrial call "Industrial Chairs" of the French National Research Agency (ANR) 2020.

Participating in / Organising local, national or international workshops

2026:

- Workshop co-chair, *"Mapping Validity in Design Research"*, Design Society, 19th International Design Conference, Dubrovnik-Cavtat, Croatia, May 18-21, 2026.
- Organising committee member, INCOSE French Chapter AFIS National Congress, January 13-15, 2026, Toulouse, France.

2025:

- Technical Program Committee Chair, Workshop IEEE Research Trends in Systems Design, 2025, Paris, France.

2024:

- Workshop co-chair, *"Design Research Quality"*, Design Society, 18th International Design Conference, Dubrovnik-Cavtat, Croatia, May 20-24, 2024.
- Workshop speaker, *"Co-design of a community-based ecosystem to improve validation practices in engineering research"*, S.mart Special Interest Group in Industry 4.0, Carry le Rouet, France, April 2023.
- Workshop coordinator and speaker, *"Advanced visualisation and interaction for Model-Based Systems Engineering"*, INCOSE French Chapter, Grenoble, France, March 2023.
- Workshop speaker, *"Research benchmarks in systems engineering: Application to the activities of early validation and verification"*, French-Academia forum of the INCOSE French Chapter, Toulouse, France, December 2022.
- Organiser of the monthly scientific seminars and the annual scientific writing workshop for the integrated design research department (3 research teams) of G-SCOP laboratory since 2018.

Local or external committee for associate and full professor selection

- External committee member, associate professor selection, Arts et Métiers ParisTech, France, 2026
- External committee member, full professor selection, IMT Mines Alès, France, 2025
- External committee member, associate professor selection, Arts et Métiers ParisTech, France, 2024
- Local committee member, associate professor selection, Grenoble INP, Grenoble, 2024
- External committee member, associate professor selection, Arts et Métiers ParisTech, France, 2023
- External committee member, associate professor selection, Arts et Métiers ParisTech, France, 2019
- External committee member, associate professor selection, Arts et Métiers ParisTech, France, 2018

Special interest groups (SIGs)

- Board member as VP of Research & Education, INCOSE French Chapter AFIS (elected in March 2025).
- Co-founder and co-chair, Design Society, Design Research Quality SIG (since 2024).
- Co-chair, INCOSE French Chapter, Local Chapter Auvergne-Rhône Alpes SIG (since 2023).
- Founder, Benchmarks for the validation of engineering design research SIG. ([Website](#))

EDUCATION

PROJECTS

- **ANR IDEX Grenoble Alpes University, Education (2025-2026)**
 - **Description:** Development of a Digital Twins authoring environment for teaching industrial engineering,
 - **Funding:** 30 000 € by Idex Grenoble Alpes University.
 - **Role:** Scientific coordinator and supervisor of 3 master's theses
- **ANR LabEx PERSYVAL Grenoble Alpes University, call for Education technological platform (2024-2025)**
 - **Description:** Development of new teaching materials and practical courses for training students on eye tracking technologies for experimental and observational approaches applied to research and industrial engineering.
 - **Funding:** 7 200 € by Labex PERSYVAL Grenoble Alpes University.
 - **Role:** Scientific coordinator and creator of teaching materials and courses

COURSES TAUGHT

2025-2026 (approx. 265 h)

- **Course director:**
 - Innovation (12h, 120 post-graduate students)
 - Systems Engineering (30h, 30 M2 post-graduate students)
 - Co-design of complex product-process systems (30h, 20 post-graduate students)
 - Conversational generative AI for industry (12h TD, 30 post-graduate students)
- **Course instructor:**
 - CAD and Product Data Management (40h, 75 post-graduate students)
 - Systems Engineering (52h, 120 post-graduate students)
 - Research Design in Industrial Engineering (22h, 30 post-graduate students)
 - Computer-Aided Design (12h, 120 post-graduate students)
 - Virtual and augmented reality (15h, 15 post-graduate students)

2024-2025 (258,76 h)

- **Course director:**

- Innovation (20h, 110 post-graduate students)
- Systems Engineering (50h, 120 post-graduate students)
- Systems Engineering (30h, 30 M2 post-graduate students)
- Co-design of complex product-process systems (30h, 15 post-graduate students)
- **Course instructor:**
- CAD and Product Data Management (42h, 75 post-graduate students)
- Research Design in Industrial Engineering, (16,5h, 25 post-graduate students)
- Computer-Aided Design (16,5h, 120 post-graduate students)
- Virtual and augmented reality (15h, 20 post-graduate students)

2023-2024 (approx. 298 h)

- **Course director:**
- Innovation (31h, 110 post-graduate students)
- Systems Engineering (86h, 120 post-graduate students)
- **Course instructor:**
- CAD and Product Data Management (47h, 70 post-graduate students)
- Research Design in Industrial Engineering, (21h, 35 post-graduate students)
- Computer-Aided Design (16h, 120 post-graduate students)
- Virtual and augmented reality (27h, 20 post-graduate students)
- Model-Based Systems Engineering (45h, 30 post-graduate students)

2022-2023 (316 h)

- **Course director:**
- Innovation (30,5h, 110 post-graduate students)
- Text mining (15h, 20 post-graduate students)
- Systems Engineering (82,5, 120 post-graduate students)
- **Course instructor:**
- CAD and Product Data Management (47h, 60 post-graduate students)
- Research Design in Industrial Engineering, (21h, 35 post-graduate students)
- Computer-Aided Design (32h, 120 post-graduate students)
- Virtual and augmented reality (30h, 30 post-graduate students)
- Model-Based Systems Engineering (21h, 20 post-graduate students)

2021-2022 (419 h)

- **Course director:**
- Innovation (30,5h, 110 post-graduate students)
- Text mining (15h, 30 post-graduate students)
- Systems Engineering (38h, 120 post-graduate students)
- **Course instructor:**
- CAD and Product Data Management (46h, 60 post-graduate students)
- Research Design in Industrial Engineering, (15h, 35 post-graduate students)
- Computer-Aided Design (15h, 30 post-graduate students)
- Virtual and augmented reality (27h, 30 post-graduate students)
- Knowledge engineering (7h, 30 post-graduate students)
- Model-Based Systems Engineering (9h, 35 post-graduate students)
- Systems Engineering Prototyping (22h, 120 post-graduate students)
- **Continuing professional development :**
- Systems Engineering course for Becton Dickinson Medical Pharmaceutical Systems (74,5 h)

2020-2021 (346 h)

- **Course creator :**
 - Text mining (15h, 30 post-graduate students)
- **Course director:**
 - Innovation (30,5h, 110 post-graduate students)
 - Systems Engineering (29h, 120 post-graduate students)
- **Course instructor:**
 - CAD and Product Data Management (48h, 60 post-graduate students)
 - Industrial engineering (32h, 60 post-graduate students)
 - Virtual and augmented reality (30h, 20 post-graduate students)
 - Systems Engineering Prototyping (30h, 120 post-graduate students)
 - Computer-Aided Design (15h TD, post-graduate students)
 - Model-Based Systems Engineering (10h, 30 post-graduate students)
 - Apprenticeship in the industry (15h, 20 post-graduate students)
- **Continuing professional development:**
 - Systems Engineering course for Becton Dickinson Medical Pharmaceutical Systems (50h)

2019-2020 (192 h)

- **Course co-creator :**
 - Virtual and augmented reality (25h, 25 post-graduate students)
- **Course director:**
 - Innovation (30,5h, 110 post-graduate students)
 - Systems Engineering (26h, 120 post-graduate students)
- **Course instructor:**
 - CAD and Product Data Management (50h, 60 post-graduate students)
 - Research Design in Industrial Engineering, (15h, 35 post-graduate students)
 - Computer-Aided Design (12h, 30 post-graduate students)
 - Modelling and Optimization in Product Development (19h, 25 post-graduate students)

2018-2019 (220 h)

- **Course instructor:**
 - Digital chain for industry 4.0 including extended reality, 20h for co-creation of a new course
 - Systems engineering (52h, 35 post-graduate students)
 - CAD and Product Data Management (47h, 60 post-graduate students)
 - Modelling and Optimization in Product Development (21h, 25 post-graduate students)
 - New assistant professor training program (64h)

Before 2018

- **Course instructor:**
 - 2017-2018. Computer-Aided Design and Product Data Management at *Arts & Métiers ParisTech* Engineering School, post-graduate students, 58 hours.
 - 2017-2018. Kinematics and dynamics of multibody systems at *Arts & Métiers ParisTech* Engineering School, post-graduate students, 20 hours.
 - 2015-2016. Computer-Aided Design and Product Data Management at *Arts & Métiers ParisTech* Engineering School, post-graduate students, 35 hours.
 - 2015-2016. Equation-based physical systems modelling and simulation at *Ecole Polytechnique Féminine* Engineering School, post-graduate students, 24 hours.

- 2014-2015. Equation-based physical systems modelling and simulation at *Ecole Polytechnique Féminine* Engineering School, post-graduate students, 16 hours.